SUN'S CIRCULAR PRIMITIVE-PERMUTATION CONJECTURE: ONE EXCEPTION AND A COMPLETE CLASSIFICATION

ANONYMOUS

Abstract. In 2013 Z.-W. Sun conjectured that for every finite field $\mathbb{F}_q$ with $q > 7$ and every $a_0 \in \mathbb{F}_q$, the nonzero elements of $\mathbb{F}_q$ can be arranged in a circle so that $a_0 + xy$ is primitive for every pair x, y of neighbours. A recent preprint (arXiv:2610.06289) proved this for $q > 18\,888\,871$ and for $a_0 = 0$. We show that the conjecture is false for $q = 11$, $a_0 = 2$, and that this is the only exception: for every prime power $q > 7$ and every $a_0 \in \mathbb{F}_q$ with $(q, a_0) \neq (11, 2)$ the required circular permutation exists. The proof combines a sieved form of the character-sum criterion, which leaves 510 prime powers $q \leq 92\,821$, with exhaustive coverage of their nonzero parameters. The supplied computational records report 3 472 147 successful cycles; replay regenerates the witnesses by heuristic search and checks them deterministically with separately implemented arithmetic, rather than checking retained cycle streams.

1. Introduction

For a prime power q and $a_0 \in \mathbb{F}_q$, let G_{q,a_0} be the graph on $\mathbb{F}_q^*$ in which distinct x, y are adjacent when $a_0 + xy$ is a primitive element of $\mathbb{F}_q$. A circular permutation $(a_1, \ldots, a_{q-1})$ of $\mathbb{F}_q^*$ with all $a_0 + a_i a_{i+1}$ primitive (indices mod $q - 1$) is exactly a Hamilton cycle of G_{q,a_0}.

Conjecture 1 (Sun [5, Conjecture 3.8, arXiv v9], dated 2013-09-17)**.** *For every finite field $\mathbb{F}_q$ with $q > 7$ and every $a_0 \in \mathbb{F}_q$, the graph G_{q,a_0} is Hamiltonian.*

Luo and She [4] proved this for $q > 18\,888\,871$ (all a_0) and for $a_0 = 0$ whenever $q > 4$, using the Chvátal–Erdős theorem and a character-sum discrepancy bound.

The contribution here is the explicit exception, the corrected exact classification, the overlapping-set independence estimate and computational closure of the finite remainder. The graph model, character bound and infinite tail are inherited from [4]; the sieve is classical [1, 2]. We also record a

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transferable vertex-and-edge-deletion corollary of the same discrepancy argument. Neither implementation diversity nor internal checking is external peer review.

Theorem 2. *Let $q > 7$ be a prime power and $a_0 \in \mathbb{F}_q$. Then G_{q,a_0} is Hamiltonian if and only if $(q, a_0) \neq (11, 2)$.*

Proposition 3. *$G_{11,2}$ is not Hamiltonian.*

Proof. The primitive roots modulo 11 are $2, 6, 7, 8$. Hence distinct $x, y \in \mathbb{F}_{11}^*$ are adjacent in $G_{11,2}$ if and only if $xy \in \{0, 4, 5, 6\}$, that is, $xy \in \{4, 5, 6\}$. The neighbours of 2 are 3 ($2 \cdot 3 = 6$) and 8 ($2 \cdot 8 = 16 \equiv 5$); the value $2 \cdot 2 = 4$ would give a loop. The neighbours of 9 are 3 ($27 \equiv 5$) and 8 ($72 \equiv 6$); here $9 \cdot 9 = 81 \equiv 4$ is again a loop. Thus 2 and 9 both have degree two with neighbourhood $\{3, 8\}$. A Hamilton cycle uses both edges at each vertex of degree two, forcing the 4-cycle $2, 3, 9, 8$. These four edges also exhaust both cycle incidences at 3 and 8, so none of the other six vertices can enter the cycle. This proves nonexistence without using failure of the search. □

2. A sieved Hamiltonicity criterion

Let $n = q - 1$, $\theta(m) = \varphi(m)/m$ and $W(m) = 2^{\omega(m)}$. For $m \mid n$, a nonzero z is *m-free* if $z = y^d$ with $d \mid m$ forces $d = 1$; primitive means n-free. For $X, Y \subseteq \mathbb{F}_q^*$ let $E_m(X, Y)$ be the number of ordered pairs $(x, y) \in X \times Y$ (with $x = y$ allowed) for which $a_0 + xy$ is m-free, and write $E = E_n$. Throughout this section $a_0 \neq 0$.

All multiplicative characters, including the principal character, are extended by $\chi(0) = 0$. The sets below may overlap. The useful improvement is to count directly on $I \times I$ for an independent set I: only the diagonal can contribute, giving $E(I, I) \leq |I|$. This avoids splitting I into disjoint parts before applying discrepancy.

Lemma 4 ([4, proof of Lemma 2.2])**.** *For every nonprincipal multiplicative character χ and all $X, Y \subseteq \mathbb{F}_q^*$, $\left|\sum_{x \in X, y \in Y} \chi(a_0 + xy)\right| \leq \sqrt{q|X||Y|}$.*

Lemma 5. *Let $e \mid n$ be squarefree (only the primes of e matter), let $p_1, \ldots, p_s$ be the primes dividing n but not e, and suppose $\delta = 1 - \sum_i 1/p_i > 0$. Put $\Delta = (s - 1)/\delta + 2$ if $s \geq 1$ and $\Delta = 1$ if $s = 0$. Then for all $X, Y \subseteq \mathbb{F}_q^*$,*

$$E(X, Y) \;\geq\; \theta(e)\,\delta\left(|X||Y| - W(e)\Delta\sqrt{q|X||Y|}\right).$$

Proof. For squarefree $m \mid n$ and $z \neq 0$,

$$\mathbf{1}_{m\text{-free}}(z) = \theta(m) \sum_{d \mid m} \frac{\mu(d)}{\varphi(d)} \sum_{\operatorname{ord}\chi = d} \chi(z).$$

For $z = 0$ every indicator vanishes. Hence

$$E_m(X, Y) = \theta(m)\big(|X||Y| - T_0\big) + \theta(m) \sum_{1 < d \mid m} \frac{\mu(d)}{\varphi(d)} \sum_{\operatorname{ord}\chi = d} S_\chi,$$

where $S_\chi = \sum_{x,y} \chi(a_0 + xy)$ and

$$T_0 = \#\{(x,y) : a_0 + xy = 0\} \le \min(|X|,|Y|) \le \sqrt{|X||Y|}.$$

With $\mathrm{rad}(n)$ in place of n (which does not change freeness), the pointwise inequality $\mathbf{1}_{\mathrm{prim}} \ge \sum_i \mathbf{1}_{ep_i\text{-free}} - (s-1)\mathbf{1}_{e\text{-free}}$ holds. Indeed, if z is primitive both sides equal 1. If z is not e-free, every term vanishes. If z is e-free but not primitive, then z is not ep_j-free for some j. Hence

$$E \;\ge\; \sum_i \big(E_{ep_i} - \theta(p_i)E_e\big) + \delta E_e,$$

because $\sum_i \theta(p_i) - (s-1) = \delta$. This is the standard core sieve [1, Lemma 2.3], [2, Section 4], applied with the nonnegative multiplicity of each value a_0+xy. Since $\theta(ep_i) = \theta(e)\theta(p_i)$, the principal-character and T_0 contributions cancel in $E_{ep_i} - \theta(p_i)E_e$. What remains is

$$\theta(e)\theta(p_i) \sum_{d \mid e} \frac{\mu(dp_i)}{\varphi(dp_i)} \sum_{\mathrm{ord}\,\chi = dp_i} S_\chi,$$

whose absolute value is at most $\theta(e)\theta(p_i)W(e)\sqrt{q|X||Y|}$. Likewise $E_e \ge \theta(e)\big(|X||Y| - W(e)\sqrt{q|X||Y|}\big)$. The coefficient identity is

$$\delta + \sum_i \theta(p_i) = s - 1 + 2\delta = \delta\Delta.$$

Combining, $E \ge \theta(e)\big[\delta|X||Y| - W(e)\sqrt{q|X||Y|}\,(\delta + \sum_i \theta(p_i))\big]$, and $\delta + \sum_i \theta(p_i) = \delta\Delta$. For $s = 0$ the bound for E_e is the claim. □

Theorem 6. *Let $a_0 \ne 0$. Suppose that for some $e, p_1, \dots, p_s$ as in Lemma 5, with $M = W(e)\Delta\sqrt{q}$ and $c = 1/(\theta(e)\delta)$,*

$$\varphi(n) - 1 - M \;\ge\; M + c.$$

Then G_{q,a_0} is Hamiltonian.

Proof. We use the Chvátal–Erdős theorem [3]: a graph on at least three vertices with connectivity κ at least its independence number α is Hamiltonian. *Degrees.* As y runs over $\mathbb{F}_q^*$, $a_0 + xy$ runs over $\mathbb{F}_q \setminus \{a_0\}$, so every degree is at least $\varphi(n) - 2$; the two lost values are a_0 and the loop $a_0 + x^2$. *Independence.* Take a nonempty maximum independent set I. Then $E(I,I) \le |I|$, since only diagonal pairs can contribute. Lemma 5 gives $\theta(e)\delta(|I|^2 - M|I|) \le |I|$, hence $\alpha \le M + c$. *Connectivity.* Let S be a minimum vertex cut, possibly empty, let A be a smallest component of $G - S$, and let B be the union of the other components. Since $E(A,B) = 0$, Lemma 5 gives $\sqrt{|A||B|} \le M$, and $|A| \le |B|$ by the choice of A, so $|A| \le \sqrt{|A||B|} \le M$. A vertex of A has at least $\varphi(n) - 2 - (|A| - 1)$ neighbours in S, so $\kappa \ge \varphi(n) - 1 - M$. If G is complete, then $\kappa = n - 1 \ge 1 = \alpha$. Hence $\kappa \ge \varphi(n) - 1 - M \ge M + c \ge \alpha$. □

3. A transferable deletion criterion

The following is a direct corollary of the discrepancy and Chvátal–Erdős argument, rather than a new general Hamiltonicity method.

Theorem 7. *Let G be a finite simple graph of minimum degree at least d. Let A be a symmetric zero–one matrix agreeing with its adjacency matrix off the diagonal, with arbitrary diagonal entries in $\{0,1\}$. Suppose $t>0$ and*

$$E(X,Y) := \sum_{x\in X, y\in Y} A_{xy} \geq t\big(|X||Y| - M\sqrt{|X||Y|}\big)$$

for all subsets X, Y, including overlapping ones. Delete at most r vertices, then edges with at most h deleted incidences at each remaining vertex, where r, h are nonnegative integers. If at least three vertices remain and

$$d+1-r-h \geq 2M + \frac{2h+1}{t}, \tag{1}$$

the remaining graph H is Hamiltonian.

Proof. For a nonempty independent set I of H, original off-diagonal incidences inside I number at most $h|I|$, and diagonal entries contribute at most $|I|$. Thus $t(|I|^2 - M|I|) \leq (h+1)|I|$, giving $\alpha(H) \leq M + (h+1)/t$. If H is complete the conclusion is immediate. Otherwise let S be a minimum vertex cut (possibly empty), X a smallest component of $H-S$, and Y the union of the others. Every original edge between X, Y was deleted. With $s = \sqrt{|X||Y|} > 0$, we have

$$t(s^2 - Ms) \leq E(X,Y) \leq h\min(|X|,|Y|) \leq hs.$$

Consequently $|X| \leq s \leq M + h/t$. Since $\delta(H) \geq d-r-h$,

$$\kappa(H) \geq d-r-h+1-M-h/t \geq M + (h+1)/t \geq \alpha(H).$$

Apply [3]. This also rules out disconnected H under the hypothesis. □

For G_{q,a_0} with $a_0 \neq 0$, substitute $d = \varphi(q-1)-2$, $t = \theta(e)\delta$ and $M = W(e)\Delta\sqrt{q}$. The exact sufficient test, usable without floating point, is

$$L := \varphi(q-1) - 1 - r - h - \frac{2h+1}{\theta(e)\delta} \geq 0, \qquad L^2 \geq 4W(e)^2\Delta^2 q.$$

The case $r = h = 0$ recovers Theorem 6. This supplies certified deletion budgets when the inequality has slack; it does not assert resilience for every residual field or for the exceptional graph, nor does the character argument apply to $a_0 = 0$.

4. Proof of Theorem 2

The case $a_0 = 0$ and the range $q > 18\,888\,871$ are covered by [4, Theorem 1.2]. Proposition 3 gives the exception. For the remaining $1\,205\,121$ prime powers $8 \leq q \leq 18\,888\,871$, exact computations test Theorem 6 over products of subsets of the prime divisors of n. They leave a set R of 510

fields, of maximum order 92 821: 471 prime fields and 39 extension fields. Separate implementations of the same mathematical criterion, using different enumeration routes, agree on R.

The supplied full-run records report a cycle for every nonzero parameter in these fields except $(11, 2)$. The search is heuristic; the parameter coverage and the checks on each proposed cycle are deterministic. The exact required coverage is

$$\sum_{q\in R}(q-1)-1 = 3\,472\,147 \quad \text{successful cycles,}$$

$$\sum_{q\in R}(q-1)^2-10 = 84\,645\,360\,506 \quad \text{cyclic edge occurrences.}$$

The subtractions remove the exceptional ten-vertex graph. A valid checked cycle proves existence regardless of how the search found it. Together with the preceding branches, this gives the classification. □

Branch or reduction	Evidence and scope
$a_0 = 0$, $q > 7$	Imported construction, [4, Theorem 1.2].
$q > 18\,888\,871$	Imported Hamiltonicity theorem, all parameters.
Finite nonzero-parameter range	Two exact implementations reduce 1 205 121 fields to 510.
$(q, a_0) = (11, 2)$	Forced 4-cycle proof, Proposition 3.
Other residual parameters	Regenerated witnesses and deterministic checking; required totals displayed above.
Fresh execution status	Per-field records and the completed replay receipt specify actual coverage, separately from historical records.

The supplied review reports a fresh replay of 332 fields and 647 047 cycles (4 381 293 458 cyclic edge occurrences), including all 39 extension fields; its remaining 178 prime-field jobs were not rerun. That reported review coverage must not be described as a fresh complete 510-field replay. Any later execution has its own receipt and scope.

Remark 8. The independence bound $\alpha \le M + c$ uses $I \times I$ rather than splitting I. The seed's literal bound is $2B + 1$, with $B = W(q-1)\sqrt{q}$; applying its splitting argument to our sieved discrepancy would give $2M+1$. These are different constants. Together with the sieve, the improvement lowers the largest unresolved q from 18 888 871 to 92 821.

The classification removes all finite-range uncertainty left by the imported theorem and corrects the literal conjecture. It does not supply an efficient uniform construction, classify all obstructions in other arithmetic graph families, or show that the deletion bound is sharp. Those would require further mathematics beyond this completed specialist classification.

Appendix A. Computations

`fails.c` lists the 2 693 fields failing the seed criterion $\varphi(n) \geq 3W(n)\sqrt{q}+2$. Then `sieve2.py` reduces them to 510. The separate `check_all.py` enumeration applies the same criterion directly to all prime powers in range. Both use the sign guard $L \geq 0$ before comparing L^2 with $4W(e)^2\Delta^2 q$, where $L = \varphi(n) - 1 - 1/(\theta(e)\delta)$. Agreement is implementation diversity, not an independent mathematical theorem or external peer review.

`ham.c` uses exponential/logarithm tables and randomized Pósa rotations. The deterministic `verify.c` uses separate schoolbook polynomial arithmetic, although it receives the same field parameters and modulus. It checks the characteristic, field size, permutation and every primitive adjacency. This is internal generator/checker separation, not formal verification. `exact_small.py` separately uses exhaustive dynamic programming for $q = 11, 13$.

For extension fields, write $R = \mathbb{F}_p[X]/(h)$ with monic h of degree f. This ring has $q = p^f$ elements before irreducibility is assumed. Checking $X^{q-1} = 1$ makes X a unit; checking $X^{(q-1)/\ell} \neq 1$ for every prime divisor $\ell \mid q-1$ gives exact order $q-1$. Its distinct powers exhaust all nonzero elements of R, which are consequently units. Thus R is a field and h is irreducible. Prime fields use arithmetic modulo the checked prime p.

The archive retains programs, parameter lists and execution records, not all historical cycle streams. Replay regenerates and immediately verifies those streams. From the repository root, with Python 3, SymPy 1.14.0 and a C compiler available as `cc` (or selected through `CC`), run

```
python3 code/replay.py --out ../replay-fresh-01 --workers 4
```

The output directory must not already exist. No shell pipeline is used. The driver records both generator and verifier statuses in each field's JSON record, and records Python/platform/source hashes in `receipt.json`. Successful ordinary jobs have both statuses zero. The expected $q = 11$ job has generator status zero and verifier status two, with exactly `missing_a0=2`; this exception is checked explicitly. A complete successful receipt must cover 510 fields, the exact totals above, and both residual routes. Selected-field or cycles-only diagnostics are not complete replays. The receipt reports actual execution; the command and expected totals alone do not certify that a run completed.

The versioned evidence release accompanying this manuscript supplies the replay code and records. Third-party mathematical sources are cited by their versioned identifiers below, rather than redistributed as original code or evidence. Neither a file hash nor a summary replaces a mathematical witness.

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